

YOLO Model Training – Individual Activity

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Plant Disease Detection Using YOLO11 on NVIDIA DGX A100

Objectives

The main objectives of this project were:

- To obtain and prepare a plant disease object-detection dataset.
- To use a YOLO11 model for plant disease detection.
- To train the model using an NVIDIA A100 GPU through the DGX/Kubeflow environment.
- To evaluate the trained model using validation and test datasets.
- To calculate object-detection performance metrics such as Precision, Recall, mAP@50, and mAP@50-95.
- To generate prediction images showing the detected plant classes.
- To save the trained model and training results for further use.

Dataset Description

The dataset used in this project was obtained from Roboflow Universe.

Dataset: Plant Disease Detection - <https://universe.roboflow.com/plant-disease-detection-mwyro/plant-disease-j1d2c>

Task: Object Detection

Number of classes: 29

Total images: 2,561

The dataset was divided into three subsets:

Dataset Split	Number of Images
Training	2,177
Validation	127
Testing	257
Total	2,561

Dataset Classes

The dataset contained the following 29 classes:

Apple Scab Leaf | Apple leaf | Apple rust leaf | Bell_pepper leaf | Bell_pepper leaf spot | Blueberry leaf | Cherry leaf | Corn Gray leaf spot | Corn leaf blight | Corn rust leaf | Peach leaf | Potato leaf | Potato leaf early blight | Potato leaf late blight | Raspberry leaf | Soyabean leaf | Squash Powdery mildew leaf | Strawberry leaf | Tomato Early blight leaf | Tomato Septoria leaf spot | Tomato leaf | Tomato leaf bacterial spot | Tomato leaf late blight | Tomato leaf mosaic virus | Tomato leaf yellow virus | Tomato mold leaf | Tomato two spotted spider mites leaf | grape leaf | grape leaf black rot

Hardware and Software Environment

The model was trained using the **Kubeflow environment connected to an NVIDIA DGX A100 system.**

Hardware

- GPU: NVIDIA A100-SXM4-40GB
- GPU allocation: 10 GB MIG partition
- GPU type shown during training: NVIDIA A100-SXM4-40GB MIG 0g.10gb
- GPU count: 1

Software

- Python: 3.10.12
- PyTorch: 2.2.0a0
- Ultralytics: 8.4.158
- YOLO version: YOLO11
- CUDA-enabled PyTorch environment
- Kubeflow Jupyter Notebook

The GPU was verified before training using PyTorch. CUDA was successfully detected and the NVIDIA A100 MIG GPU was available for model training.

```
PyTorch version: 2.2.0a0+81ea7a4
CUDA available: True
GPU: NVIDIA A100-SXM4-40GB MIG 0g.10gb
GPU count: 1
```

Training Configuration

Parameter	Value
Model	YOLO11n
Dataset	Plant Disease Detection
Number of classes	29
Image size	640 × 640
Batch size	4
Maximum epochs	100
Workers	4
GPU	NVIDIA A100
Pretrained	Yes
Early stopping patience	20
Training images	2,177
Validation images	127
Test images	257

The maximum number of epochs was set to **100**. However, early stopping was enabled with a patience value of 20. As a result, the training process stopped automatically after **60 epochs** because the validation performance did not show sufficient improvement.

The **best model was obtained at epoch 40**, and the corresponding best .pt checkpoint was saved.

```

60 epochs completed in 0.467 hours.
Optimizer stripped from /workspace/plant-disease-results/plant_disease_yolo11n/weights/last.pt, 5.5MB
Optimizer stripped from /workspace/plant-disease-results/plant_disease_yolo11n/weights/best.pt, 5.5MB

Validating /workspace/plant-disease-results/plant_disease_yolo11n/weights/best.pt...
Ultralytics 8.4.158 Python-3.10.12 torch-2.2.0a0+81ea7a4 CUDA:0 (NVIDIA A100-SXM4-40GB MIG 0g.10gb, 9984MiB)
YOLO11n summary (fused): 100 layers, 2,587,807 parameters, 0 gradients, 6.4 GFLOPs
YOLO11n summary (fused): 100 layers, 2,587,807 parameters, 0 gradients, 6.4 GFLOPs
Class      Images  Instances  Box(P  R  mAP50  mAP50-95) 100% 16/16 20.3it/s 0.8s0.1s
all        127      512      0.46   0.561 0.573 0.405
Apple Scab Leaf 4 7 0.762 0.857 0.913 0.623
Apple leaf 6 10 0.783 1 0.895 0.636
Apple rust leaf 5 5 0.305 0.183 0.145 0.0793
Bell pepper leaf 4 30 0.298 0.7 0.622 0.407
Bell pepper leaf spot 1 9 0.338 0.556 0.536 0.291
Blueberry leaf 8 73 0.327 0.808 0.646 0.387
Cherry leaf 2 14 0.669 0.286 0.654 0.455
Corn Gray leaf spot 3 3 0.465 0.333 0.552 0.505
Corn leaf blight 7 11 0.34 0.545 0.375 0.243
Corn rust leaf 7 8 0.394 0.875 0.877 0.681
Peach leaf 6 44 0.673 0.886 0.83 0.588
Potato leaf early blight 6 25 0.251 0.84 0.327 0.179
Potato leaf late blight 4 4 0 0 0.0989 0.0652
Raspberry leaf 5 30 0.818 0.633 0.795 0.55
Soybean leaf 4 13 0.513 1 0.986 0.792
Squash Powdery mildew leaf 9 29 25 0.467 0.931 0.48 0.418 0.209
Strawberry leaf 5 29 0.467 0.931 0.753 0.597
Tomato Early blight leaf 2 7 0 0 0.193 0.106
Tomato Septoria leaf spot 10 22 0.755 0.545 0.776 0.514
Tomato leaf 6 43 0.469 0.698 0.559 0.292
Tomato leaf bacterial spot 6 5 9 0.115 0.167 0.105 0.0866
Tomato leaf late blight 6 9 0.427 0.667 0.59 0.516
Tomato leaf mosaic virus 2 8 0.254 0.75 0.466 0.327
Tomato leaf yellow virus 5 57 0.417 0.491 0.452 0.253
Tomato mold leaf 5 19 0.821 0.158 0.347 0.258
grape leaf black rot 1 1 0.854 1 0.995 0.895

Speed: 0.4ms preprocess, 2.1ms inference, 0.0ms loss, 0.9ms postprocess per image
Results saved to /workspace/plant-disease-results/plant_disease_yolo11n

```

Model Evaluation

After training, the saved best .pt model was evaluated on the test dataset containing **257 images and 790 annotated instances**.

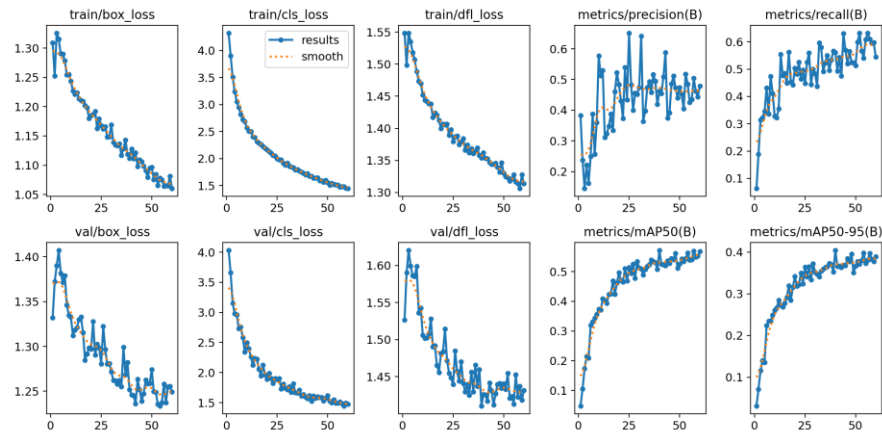
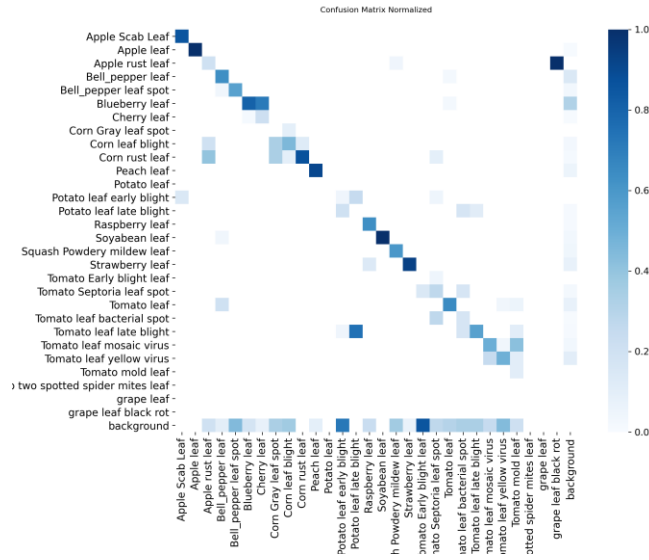
The final test results were:

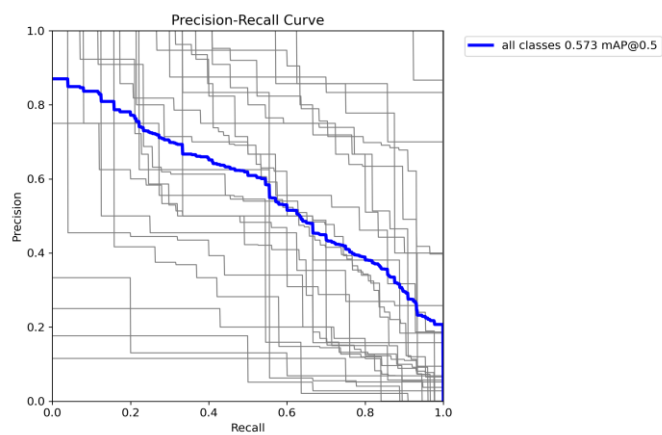
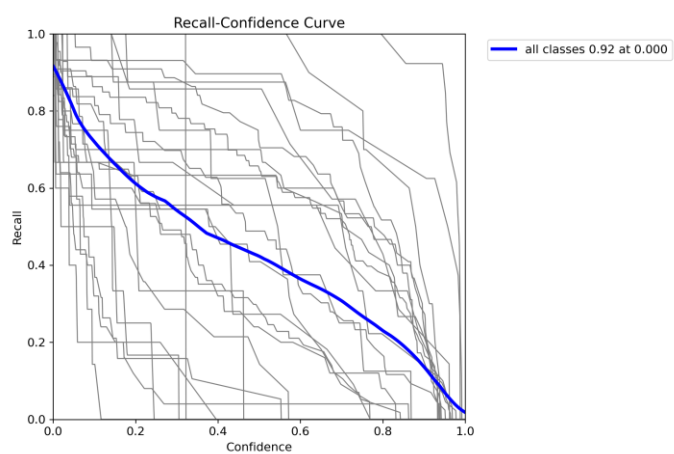
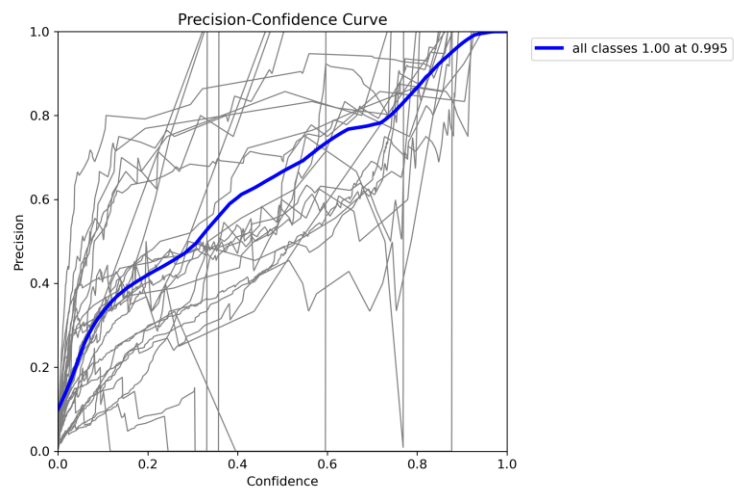
Metric	Result
Precision	50.7%
Recall	56.2%
mAP@50	54.2%
mAP@50-95	37.9%

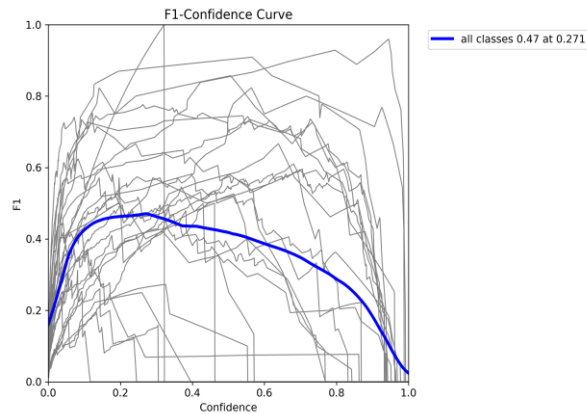
Confusion Matrix and Training Graphs

Several graphs were generated during training and evaluation, including:

- Training and validation loss graphs
- Precision curve
- Recall curve
- F1-score curve
- Precision-Recall curve
- Confusion matrix
- Normalized confusion matrix
- Label distribution visualization







Prediction Results

After completing the evaluation, the trained best .pt model was used to generate predictions on the test images.

The predictions were generated with:

- Image size: 640 × 640
- Confidence threshold: 0.25
- GPU: NVIDIA A100

The resulting images contain bounding boxes and predicted class names produced by the YOLO11 model.



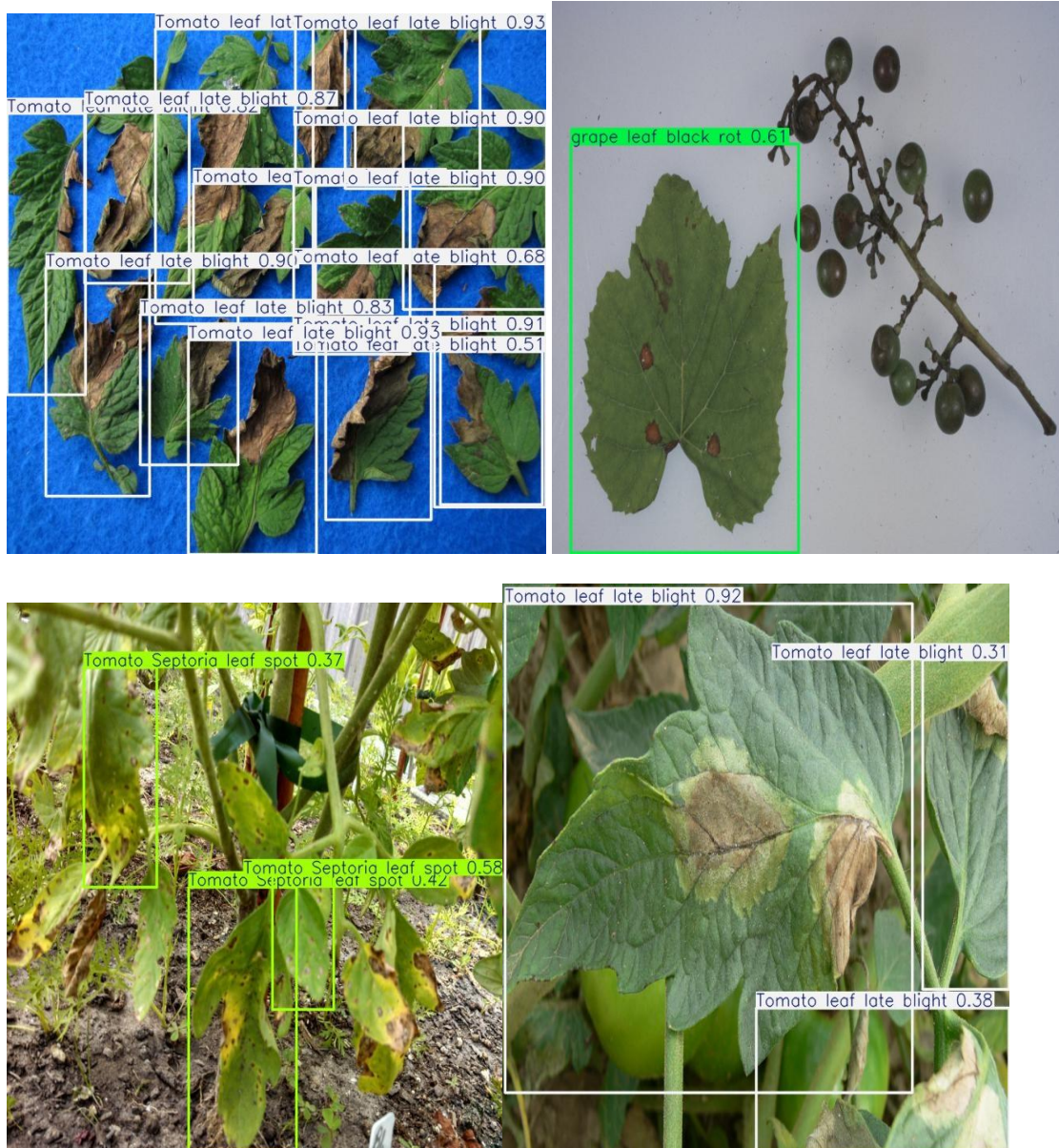


Figure: YOLO11n predictions on test images showing detected plant/leaf classes with bounding boxes and confidence scores.

Model Files and Output

The main output files generated during the project included:

- best.pt – best-performing trained model
- last.pt – final training checkpoint
- results.csv – training metrics for each epoch
- results.png – training and validation graphs
- confusion_matrix.png – confusion matrix

- `confusion_matrix_normalized.png` – normalized confusion matrix
- `BoxP_curve.png` – precision curve
- `BoxR_curve.png` – recall curve
- `BoxF1_curve.png` – F1 curve
- `BoxPR_curve.png` – Precision-Recall curve
- Test evaluation results
- Test prediction images
- YOLO training notebook (`.ipynb`)

These files provide evidence of the complete training and evaluation process.

Conclusion

In this project, a **YOLO11n object-detection model** was trained to detect and classify plant-related diseases and leaf categories using a Roboflow dataset containing **2,561 images across 29 classes**.

The training was successfully performed using an **NVIDIA DGX A100 GPU through Kubeflow**. The model was trained for a maximum of 100 epochs with early stopping, and the training process ended after 60 epochs. The best model was obtained at epoch 40.

On the independent test dataset, the model achieved:

- **Precision: 50.7%**
- **Recall: 56.2%**
- **mAP@50: 54.2%**
- **mAP@50-95: 37.9%**

The project demonstrated the complete workflow of an object-detection application, from dataset preparation and GPU configuration to model training, evaluation, and prediction generation.

The trained model and associated outputs can be used as a basis for further improvement, such as increasing the dataset size, improving class balance, applying additional data augmentation, tuning training parameters, or using a larger YOLO model.

SCREENSHOTS

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docker-examples 3 years ago
examples 3 years ago
Plant_Disease_YOLO11_Sub... 5 minutes ago
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plant-disease-results 25 minutes ago
tutorials 3 years ago
weights an hour ago
NVIDIA_Deep_Learning_Co... 3 years ago
Plant_Disease_YOLO11_Sub... 4 minutes ago
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Untitled.ipynb 3 minutes ago
yolo11n.pt an hour ago

Launcher
x Untitled.ipynb
Python 3 (pykernel)

[8]: dataset = version.download("yolo11")
print("Downloaded to:", dataset.location)
Downloaded to: /workspace/plant-disease-1

[9]: import os

print(os.listdir(dataset.location))
['valid', 'data.yaml', 'train', 'test', 'README.dataset.txt', 'README.roboflow.txt']

[10]: import os
import yaml

dataset_path = "/workspace/plant-disease-1"

with open(os.path.join(dataset_path, "data.yaml"), "r") as f:
    data = yaml.safe_load(f)

print("Dataset path:", dataset_path)
print("Classes:", data["names"])
print("Number of classes:", data["nc"])

for split in ["train", "valid", "test"]:
    image_dir = os.path.join(dataset_path, split, "images")
    label_dir = os.path.join(dataset_path, split, "labels")

    images = os.listdir(image_dir)
    labels = os.listdir(label_dir)

    print(f"\n{split}:")
    print("  Images:", len(images))
    print("  Labels:", len(labels))

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plant-disease-results 26 minutes ago
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weights an hour ago
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Plant_Disease_YOLO11_Sub... 5 minutes ago
README.md 3 years ago
Untitled.ipynb 3 minutes ago
yolo11n.pt an hour ago

Launcher
x Untitled.ipynb
Python 3 (pykernel)

[11]: !nvidia-smi

Tue Sep 22 06:20:27 2026
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NVIDIA-SMI 470.223.02 Driver Version: 470.223.02 CUDA Version: 12.3
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+---+---+---+---+---+---+---+---+---+---+---+---+---+---+---+---+
| GPU   Name              Persistence-M | Bus-Id  Disp.A | Volatile Uncorr. ECC |
| Fan  Temp  Perf    Pwr:Usage/Cap |  Memory-Usage | GPU-Util  Compute M. |
|=====================================+-----+-----+
| 0  NVIDIA A100-SXM...  On          | 00000000:08:00:0 Off |   N/A   N/A   Default On
| N/A   28C    P0      55W / 400W | 1024MiB / 65536MiB |   0%      Default  |
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MIG devices:
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| GPU   GI   CI   MIG ID  | Memory-Usage | Vol  | Shared |
| ID   ID   ID   Dev  | BARR1-Usage | SM   | Unc   | CE  ENC  DEC  OFA  JPG |
|=====+-----+-----+
| 0    6    0    0      | 6MiB / 9984MiB | 28   | 0     | 2   0   1   0   0   |
| 0MiB / 16383MiB |
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Processes:
+---+---+---+---+---+---+---+---+---+---+---+---+---+---+---+---+
| GPU   GI   CI   PID  Type  Process name                        | GPU Memory |
| ID   ID   ID   ID   |              | Usage |
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| No running processes found |
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0 1 Python 3 (pykernel) | Idle
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plant-disease-1 an hour ago
plant-disease-results 27 minutes ago
tutorials 3 years ago
weights an hour ago
NVIDIA_Deep_Learning_Co... 3 years ago
Plant_Disease_YOLO11_Sub... 6 minutes ago
README.md 3 years ago
Untitled.ipynb 4 minutes ago
yolo11n.pt an hour ago

Launcher
x Untitled.ipynb
Python 3 (pykernel)

[7]: from ultralytics import YOLO

# Load pretrained YOLO11 Nano
model = YOLO("yolo11n.pt")

# Full training
results = model.train(
    data="/workspace/plant-disease-1/data.yaml",
    epochs=100,
    imgsz=640,
    batch=4,
    device=0,
    workers=4,
    project="/workspace/plant-disease-results",
    name="plant_disease_yolo11n",
    exist_ok=True,
    pretrained=True,
    patience=20,
    plots=True,
    save=True
)

Ultralytics 8.4.158 Python-3.10.12 torch-2.2.0a0+81ea744 CUDA:0 (NVIDIA A100-SXM4-40GB MIG 0g.10gb, 9984MiB)
engine/trainer: agnostic_nms=False, amp=True, angle=1.0, augment=False, auto_augment=RandAugment, batch=4, bgr=False, box=7.5, cache=False, cfg=None,
channels_last=None, classes=None, close_mosaic=10, cls=0.5, cls_pw=0.0, cls_remap=True, compile=False, conf=None, copy_paste=0.0, copy_paste_mode=flip,
cos_lr=False, cutmix=0.0, data=/workspace/plant-disease-1/data.yaml, degrees=0.0, deterministic=True, device=0, dfl=1.5, dgrad=0.5, dis=6.0,
distill_model=None, diam=1.0, diag=1.0, dnn=False, dropout=0.0, dynamic=False, embed=None, epochs=100, erasing=0.4, exist_ok=True, flipplr=0.5,
flipud=0.5, flipudv=0.0, format=torchscript, fraction=1.0, freeze=None, hsv_h=0.015, hsv_s=0.7, hsv_v=0.4, imgsz=640, iou=0.7, keras=False, kobj=1.0,
line_width=None, lr=0.01, lr_f=0.01, mask_ratio=4, max_det=100, mosaic=0.0, mode=train, model=yolo11n.pt, momentum=0.937, mosaic=1.0, multi_scale=0.0,
name=plant_disease_yolo11n, nbs=64, nms=None, opset=None, optimizer=False, optimizer=auto, overlap_mask=True, patience=20, perspective=0.0,
plots=True, pose=12.0, pretrained=True, profile=False, project=/workspace/plant-disease-results, quantize=None, rect=False, resume=False,
retina_masks=False, rle=1.0, save=True, save_conf=False, save_crop=False, save_dir=/workspace/plant-disease-results/plant_disease_yolo11n,
save_frames=False, save_json=False, save_period=1, save_txt=False, scale=0.5, seed=0, shear=0.0, show=False, show_boxes=True, show_conf=True,
show_labels=True, simplify=True, single_cls=False, source=None, split=val, stream_buffer=False, task=detect, time=None, tracker=tracktrack.yaml
```

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Name Last Modified
docker-examples 3 years ago
examples 3 years ago
Plant_Disease_YOLO11_Sub... 7 minutes ago
plant-disease-1 an hour ago
plant-disease-results 27 minutes ago
tutorials 3 years ago
weights an hour ago
NVIDIA_Deep_Learning.Co... 3 years ago
Plant_Disease_YOLO11_Sub... 7 minutes ago
README.md 3 years ago
Untitled.ipynb 5 minutes ago
yolo11n.pt an hour ago

Launcher
Python 3 (pykernel)
60 epochs completed in 0.467 hours.
Optimizer stripped from /workspace/plant-disease-results/plant_disease_yolo11n/weights/last.pt, 5.5MB
Optimizer stripped from /workspace/plant-disease-results/plant_disease_yolo11n/weights/best.pt, 5.5MB
Validating /workspace/plant-disease-results/plant_disease_yolo11n/weights/best.pt...
Ultralytics 8.4.150 Python-3.10.12 torch-2.2.0a0+81ea7a4 CUDA:0 (NVIDIA A100-SXM4-40GB MIG 0g.10gb, 9984MiB)
YOLO11 summary (fused): 180 layers, 2,587,807 parameters, 0 gradients, 6.4 GFLOPs
Class Images Instances Box(P R mAP50 mAP50-95): 100% 16/16 20.31t/s 0.8x0.1s
all 127 512 0.46 0.561 0.573 0.405
Apple Scab Leaf 4 7 0.762 0.857 0.913 0.623
Apple leaf 6 10 0.783 1 0.895 0.636
Apple rust leaf 5 5 0.305 0.183 0.145 0.0793
Bell_pepper leaf 4 30 0.298 0.7 0.622 0.407
Bell_pepper leaf spot 1 9 0.338 0.556 0.536 0.291
Blueberry leaf 8 73 0.327 0.808 0.646 0.387
Cherry leaf 2 14 0.660 0.286 0.654 0.455
Corn Gray leaf spot 3 3 0.465 0.333 0.552 0.505
Corn leaf blight 7 11 0.34 0.545 0.375 0.243
Corn rust leaf 7 8 0.394 0.875 0.877 0.681
Peach leaf 6 44 0.673 0.886 0.83 0.588
Potato leaf early blight 6 25 0.251 0.04 0.327 0.179
Potato leaf late blight 4 4 0 0 0.0989 0.0652
Raspberry leaf 5 30 0.818 0.633 0.795 0.55
Soybean leaf 4 13 0.513 1 0.986 0.792
Squash Powdery mildew leaf 5 9 25 0.448 0.48 0.418 0.209
Strawberry leaf 2 29 0.467 0.931 0.753 0.597
Tomato Early blight leaf 2 7 0 0 0.193 0.106
Tomato Septoria leaf spot 10 22 0.755 0.545 0.776 0.514
Tomato leaf 6 43 0.469 0.698 0.559 0.292
Tomato leaf bacterial spot 5 6 0.115 0.167 0.105 0.0066
Tomato leaf late blight 6 9 0.427 0.667 0.59 0.516
Tomato leaf mosaic virus 2 8 0.254 0.75 0.466 0.327
Tomato leaf yellow virus 5 57 0.417 0.491 0.452 0.253
Tomato mold leaf 5 19 0.821 0.158 0.347 0.258
grape leaf black rot 1 1 0.854 1 0.995 0.895
Speed: 0.4ms preprocess, 2.1ms inference, 0.0ms loss, 0.9ms postprocess per image
Results saved to /workspace/plant-disease-results/plant_disease_yolo11n
```

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Name Last Modified
docker-examples 3 years ago
examples 3 years ago
Plant_Disease_YOLO11_Sub... 8 minutes ago
plant-disease-1 an hour ago
plant-disease-results 28 minutes ago
tutorials 3 years ago
weights an hour ago
NVIDIA_Deep_Learning.Co... 3 years ago
Plant_Disease_YOLO11_Sub... 8 minutes ago
README.md 3 years ago
Untitled.ipynb 6 minutes ago
yolo11n.pt an hour ago

Launcher
Python 3 (pykernel)
[9]: from ultralytics import YOLO

model = YOLO(
    "/workspace/plant-disease-results/plant_disease_yolo11n/weights/best.pt"
)

predictions = model.predict(
    sources="/workspace/plant-disease-1/test/images",
    imgs=640,
    batch=4,
    conf=0.25,
    device=0,
    save=True,
    projects="/workspace/plant-disease-results",
    name="test_predictions",
    exist_ok=True
)

print("Prediction images saved.")

image 1/257 /workspace/plant-disease-1/test/images/img1001.jpg.rf.63e2e8641859635ec8bb5b0cfdfe2b.jpg: 640x640 1 Corn rust leaf, 2.4ms
image 2/257 /workspace/plant-disease-1/test/images/img1004.jpg.rf.1dc35081d3ee08ec1020c7bbcb5ef23.jpg: 640x640 1 Apple leaf, 2.4ms
image 3/257 /workspace/plant-disease-1/test/images/img1015.jpg.rf.1a531fb2f9970e0b01cb779a67db0f8d.jpg: 640x640 1 Corn leaf blight, 2.4ms
image 4/257 /workspace/plant-disease-1/test/images/img1021.jpg.rf.910f47c5feb0497c9f8a474211081bfc.jpg: 640x640 3 Tomato leaf late blights, 2.4ms
image 5/257 /workspace/plant-disease-1/test/images/img1024.jpg.rf.e8a3c69f111acbad8a036fc9b178e845.jpg: 640x640 (no detections), 2.5ms
image 6/257 /workspace/plant-disease-1/test/images/img103.jpg.rf.41fd85890c52a530b37479a68975142a.jpg: 640x640 1 Potato leaf early blight, 1 To
mato Septoria leaf spot, 2.5ms
image 7/257 /workspace/plant-disease-1/test/images/img1070.jpg.rf.6365458b5830ea3c11cf49592580d194.jpg: 640x640 3 Tomato Septoria leaf spots,
2.5ms
image 8/257 /workspace/plant-disease-1/test/images/img1076.jpg.rf.3dcfe2f7b25208eeefac2e2ab924ae5.jpg: 640x640 12 Blueberry leaves, 11 Soybean
leaves, 2.5ms
image 9/257 /workspace/plant-disease-1/test/images/img1086.jpg.rf.d340faa99f1229b55feb0e82d1ba32b4.jpg: 640x640 1 Apple rust leaf, 2 grape leaf
black rots, 2.5ms
image 10/257 /workspace/plant-disease-1/test/images/img1095.jpg.rf.ba7d7c1cb0fc5115e9d7c19367dc1d16.jpg: 640x640 1 Squash Powdery mildew leaf.
```

```
< -> 192.168.12.131380/notebook/s-prachi-2024csai004/assignment2/lab?
LeetCode - The Wor... CodeChef Codio | Tracker GitHub LinkedIn YouTube Gmail ChatGPT DSA patterns Cheat... WhatsApp Top 65+ Angular Int... C# Progress Sheet ...

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Name Last Modified
docker-examples 3 years ago
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Plant_Disease_YOLO11_Sub... 8 minutes ago
plant-disease-1 an hour ago
plant-disease-results 29 minutes ago
tutorials 3 years ago
weights an hour ago
NVIDIA_Deep_Learning.Co... 3 years ago

Launcher
Python 3 (pykernel)
[10]: import os

path = "/workspace/plant-disease-results/test_predictions"
print(os.listdir(path)[:20])

['img273.jpg.rf.d4826ec96f1fcd617adfed394963803.jpg', 'img1589.jpg.rf.c756cc23fd8a7769ea2d376d2e9fa13.jpg', 'img1368.jpg.rf.62306b84b0db8c2e2
0eacdf0a8595b9.jpg', 'img1806.jpg.rf.d1836a8d9d68b1ec16f395fc24268cc1.jpg', 'img1535.jpg.rf.7b2062e10ae74d426da37bf73a8fd6c3.jpg', 'img1551.jp
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PyTorch version: 2.2.0a0+81ea7a4
CUDA available: True
GPU: NVIDIA A100-SXM4-40GB MIG 0g.10gb
GPU count: 1

